AWESOME CYBER SECURITY HEAVEN — REALM DOSSIER · 2026 EDITION

The Sentinel Ops

Blue Team, SOC & DFIR Tools

97	280,705	69%	50
FREE TOOLS	COMMUNITY STARS	ON GITHUB ACTIVE	ACTIVE IN 2026

97 tools · Part of Awesome Cyber Security Heaven — 1435 curated free tools across 14 realms · Data verified 2026-11-06 (re-verified monthly)

CURATOR'S NOTES

The largest realm — detection engineering, ATT&CK-aligned datasets, SIEM labs and DFIR toolkits. The fingerprinting standards (JA3, HASSH) live here: archived upstream, still canonical, still cited in nearly every detection paper.

MAINTENANCE SNAPSHOT (2026-11-06)

- **Active in the last 30 days:** 49 of 71 tracked GitHub repos (69%)
- **Quiet classics (no push in 12+ months, still listed):** none
- **Earliest entries:** The Sleuth Kit (2011), Zeek (2012), Arkime (2012)

FLAGSHIP PICKS — THE HEAVYWEIGHTS

TOOL	★ STARS	CREATED	LAST PUSH	WHAT IT DOES
osquery	23.7k	2014	2026-10-31 ✅	SQL powered operating system instrumentation, monitoring, and analytics.
Wazuh	17.5k	2015	2026-11-01 ✅	Wazuh - The Open Source Security Platform. Unified XDR and SIEM protection for endpoints and cloud workloads.
Atomic Red Team	12.7k	2017	2026-10-31 ✅	Small and highly portable detection tests based on MITRE's ATT&CK.
Sigma	11.4k	2016	2026-11-01 ✅	Main Sigma Rule Repository
awesome-honeypots	10.6k	2015	2026-06-01	an awesome list of honeypot resources
YARA	10k	2012	2026-10-27 ✅	The pattern matching swiss knife

Full inventory — 97 curated tools · ranked by GitHub adoption · 2026-11-06 census · non-GitHub tools listed without stars

TOOL	★ STARS	CREATED	LAST PUSH	LICENSE / TIER
osquery	23.7k	2014	2026-10-31 ✅	— operate endpoints as a SQL database: ask security questions, get rows back. Used by IT-security teams querying endpoints with SQL; the open-source foundation Facebook built for fleet visibility.
Wazuh	17.5k	2015	2026-11-01 ✅	— open-source XDR/SIEM: FIM, log analysis, vulnerability detection and compliance reporting in one free platform. Used by SOC teams running open XDR and SIEM; the free alternative to Splunk, QRadar and commercial EDR stacks. (GPL-2.0)
Atomic Red Team	12.7k	2017	2026-10-31 ✅	— library of executable detection tests mapped to MITRE ATT&CK; prove your defenses fire. Used by purple teams testing detections per ATT&CK technique; the open-source alternative to commercial breach-attack-simulation platforms.
Sigma	11.4k	2016	2026-11-01 ✅	— vendor-neutral SIEM detection rules — "Snort for logs", convertible to Splunk/Sentinel/Elastic. Used by detection engineers writing portable detection rules; the vendor-neutral alternative to proprietary SIEM rule languages. (DRL 1.1 rules)
awesome-honeypots	10.6k	2015	2026-06-01	— the curated honeypot directory: deception tech from low to high interaction. Used by deception researchers picking deployable honeypots. (curated list)
YARA	10k	2012	2026-10-27 ✅	— the pattern-matching language for malware identification — if you write detections, you write YARA. Used by malware analysts everywhere to classify samples; the pattern-matching standard commercial AV also uses. (BSD-3)
Awesome Incident Response	9.5k	2015	2026-10-13 ✅	— the curated IR toolkit index used across the industry. Used by DFIR professionals assembling their toolkit. (Apache-2.0, curated list)

FLARE VM	9.3k	2017	2026-10-07 ✅	— Windows security distribution bundling hundreds of curated tools for malware analysis, reverse engineering, and incident response. Used by analysts and DFIR teams; a comprehensive ready-to-deploy analysis workstation. (Apache-2.0)
Maltrail	8.7k	2014	2026-11-05 ✅	— Malicious traffic detection system using publicly available blocklists and custom blacklists. Used by network defenders and SOC analysts; a simple sensor for spotting known-bad traffic. (MIT)
ntopng	8.3k	2015	2026-10-31 ✅	— high-speed web-based traffic monitoring: flows, top talkers and protocols with long historical views. Used by network engineers observing traffic in real time; the open-source traffic-analysis alternative to commercial NPM tools. (GPL-3.0)
Zeek	8.2k	2012	2026-10-31 ✅	— the network analysis framework that turns raw traffic into structured, queryable intelligence. Used by NSM teams producing rich network telemetry; the framework most SOC pipelines are built around. (BSD-3)
Arkime	7.6k	2012	2026-10-30 ✅	— full-packet capture plus indexed search at scale (formerly Moloch); PCAP you can actually query. Used by NSM teams capturing and searching full packets at scale; the open-source alternative to commercial full-packet-capture appliances.
Suricata	6.8k	2012	2026-10-31 ✅	— high-performance IDS/IPS/NSM engine with full protocol logging and Eve JSON output. Used by network defenders running IDS/IPS at line rate; the open-source engine behind countless commercial and gov deployments. (GPL-2.0)
capa	6.3k	2020	2026-10-31 ✅	— identifies what malware actually does: maps capabilities and ATT&CK techniques straight from binaries. Used by malware analysts determining what binaries can do; the open-source capability engine from Mandiant research.
pspy	6.2k	2018	2026-03-01	— watch process and file events on any Linux box without root — catch cron-based persistence in the act. Used by pentesters and CTF players watching process activity without root; the standard privesc enumeration tool. (GPL-3.0)
ThreatMapper	5.3k	2020	2026-06-01	— Deepfence's open runtime threat scanner: finds and ranks exploitable vulnerabilities in cloud and container workloads. Used by cloud security teams mapping runtime risks; Deepfence's open-source alternative to commercial CNAPPs.
GRR Rapid Response	5.1k	2013	2026-10-01 ✅	— Incident response framework focused on remote live forensics with a Python agent on assets and server infrastructure. Used by IR teams; a scalable open-source forensic collection platform. (Apache-2.0)
Security Onion	5k	2018	2026-10-30 ✅	— the free SOC-in-a-box distro: Suricata, Zeek, Elastic, dashboards and detection pipelines. Used by SOC and DFIR teams as a complete network-security-monitoring distro; the free alternative to commercial NSM suites. (Elastic License 2.0)
Threat Hunter Playbook	4.7k	2017	2026-01-12	— open hunting playbooks as code: hypotheses mapped to data sources and detections. Used by threat hunters learning detection strategy; the open knowledge base for hunting methodology. (MIT)
Volatility 3	4.6k	2014	2026-10-12 ✅	— memory forensics framework — RAM malware has nowhere to hide. Used by DFIR analysts extracting artifacts from memory images; the framework that defined open memory forensics.
BlueTeam-Tools	4.6k	2023	2026-08-23	— community-fed directory of 70+ blue-team tools and operational resources across 13 categories plus practical defensive tradecraft tips. Used by SOC builders assembling their defensive stack; the blue-team counterpart to red-team toolkit lists. (curated list)
Velociraptor	4.4k	2018	2026-10-31 ✅	— endpoint visibility and DFIR at scale: hunt across thousands of machines with one query language. Used by DFIR teams hunting across thousands of endpoints; the open-source alternative to commercial endpoint-forensics platforms. (AGPL-3.0)
Awesome YARA	4.3k	2017	2026-06-15	— curated YARA rules, tooling and research for malware hunting. Used by malware analysts collecting detection rules. (CC-BY-4.0, curated list)
FLOSS	4.2k	2016	2026-11-01 ✅	— Mandiant's FLARE Obfuscated String Solver that automatically extracts and deobfuscates strings from malware binaries. Used by malware analysts and reverse engineers; a leading automated string extraction tool. (Apache-2.0)
HELK (Hunting ELK)	3.9k	2017	2024-06-01 ⏳	— All-in-one free threat hunting stack built on Elasticsearch, Logstash, Kafka, and Kibana with Jupyter Notebook integrations. Used by threat hunters; a comprehensive open hunting platform. (GPL-3.0)

CAPEv2	3.6k	2019	2026-10-31 ✅	— the maintained Cuckoo sandbox: automated malware detonation, config extraction and reporting. Used by malware analysts running automated sandbox detonations; the open-source alternative to commercial sandboxes like JoeSandbox. (GPL-3.0)
Snort 3	3.5k	2014	2026-10-21 ✅	— the next generation of the classic open-source IDS/IPS, rewritten in C++ with multi-threading. Used by network defenders running the classic open IDS; the engine that defined open-source intrusion detection. (GPL-2.0)
Timesketch	3.4k	2014	2026-11-01 ✅	— Google's collaborative forensic timeline analysis for multi-investigator cases. Used by DFIR analysts building collaborative forensic timelines; the open-source alternative to commercial timeline-analysis tools.
oletools	3.4k	2016	2026-02-14 ⏳	— Python tools for analyzing Microsoft OLE2 files including Office documents and MSI files, targeting malicious macros and embedded objects. Used by malware analysts and document forensicators; an essential Office analysis toolkit. (NOASSERTION)
Autopsy	3.4k	2011	2026-06-20 ⏳	— Open-source disk forensics platform with a GUI for analyzing E01, raw, and VHD images plus carving and keyword modules. Used by investigators; a widely adopted free forensic suite.
sysmon-modular	3.2k	2018	2026-10-14 ✅	— Swiss-army Sysmon configuration: per-ATT&CK-technique logging modules you compose yourself. Used by detection engineers tailoring Sysmon per event type; the community config alternative to fixed baselines.
The Sleuth Kit	3.2k	2011	2026-10-27 ✅	— the forensic library behind half the DFIR tools you know: disks, filesystems and timelines. Used by DFIR analysts examining disk images and file systems; the open-source foundation under commercial forensic suites. (CPL-1.0)
Elastic detection-rules	2.9k	2020	2026-10-31 ✅	— the detection corpus for Elastic Security: hundreds of maintained rules mapped to MITRE. Used by SOC teams deploying detection content; the open-source ruleset behind Elastic Security. (Elastic-2.0)
Fibratus	2.6k	2016	2026-10-30 ✅	— Windows kernel-event tracing with YARA filtering; a Swiss army knife for endpoint visibility. Used by defenders tracing Windows kernel activity for threats; the open-source kernel-observability tool. (Apache-2.0)
Malcolm	2.6k	2019	2026-11-03 ✅	— Network traffic analysis tool suite for full packet capture artifacts, Zeek logs, and Suricata alerts, published by CISA. Used by blue teams and incident responders; a deployable platform for deep traffic visibility. (Apache-2.0)
Shuffle	2.5k	2020	2026-10-27 ✅	— self-hosted SOAR automation: connect your TIP, SIEM and responders in drag-and-drop workflows. Used by SOC teams automating playbooks; the open-source SOAR alternative to Splunk SOAR and XSOAR. (AGPL-3.0)
DeepBlueCLI	2.4k	2016	2023-10-14 ⏳	— PowerShell module for hunt teaming via Windows Event logs, enabling rapid triage of security events from PowerShell. Used by incident responders; a fast free event log triage tool. (GPL-3.0)
Kismet	2.3k	2016	2026-11-01 ✅	— passive wireless detection layer: discovers Wi-Fi and BLE devices, rogue access points and wardrives. Used by wireless security auditors monitoring WiFi and Bluetooth; the standard open wireless detection platform. (GPL-2.0)
Adalanche	2.2k	2020	2026-10-29 ✅	— Active Directory attack-graph visualizer and explorer: ingest AD objects and see the real paths to Domain Admin. Used by blue teams auditing ACLs and red teams planning paths; the go-to open-source AD attack-path analyzer. (AGPL-3.0)
Sysmon for Linux	2.2k	2020	2026-10-28 ✅	— the Windows-trusted system monitor ported to Linux. Used by defenders unifying endpoint telemetry across OSes; Microsoft's cross-platform extension of Sysmon. (MIT)
FakeNet-NG	2.2k	2016	2026-10-30 ✅	— Dynamic network analysis tool by Mandiant simulating internet services to intercept and redirect malware C2 traffic. Used by malware analysts and sandbox operators; enables safe analysis of network behavior. (Apache-2.0)
Plaso / log2timeline	2.2k	2014	2026-11-02 ✅	— Super-timeline engine that extracts timestamps from dozens of forensic sources into a unified chronology. Used by forensic analysts; the standard open-source timeline builder. (Apache-2.0)
Nagios Core	2.1k	2014	2026-10-27 ✅	— the classic infrastructure monitor: availability checks that surface outages and anomalies early. Used by ops teams monitoring infrastructure for decades; the classic open-source monitoring platform. (GPL-2.0)
Firmadyne	2.1k	2016	2024-07-21 ⏳	— Automated firmware emulation platform for Linux-based IoT devices enabling dynamic analysis in QEMU. Used by firmware analysts; a foundational emulation tool. (MIT)

FIR	2k	2015	2026-09-14	— Fast Incident Response from Société Générale: ticketing built for IR workflows. Used by incident-response teams managing cases; the open-source IR platform for teams without commercial case tools. (GPL-3.0)
msticpy	2k	2019	2026-10-29 ✅	— SecOps analysis library from Microsoft: pivots, timelines and hunting helpers for Jupyter. Used by defenders hunting in Microsoft 365 and Sentinel data; Microsoft's open hunting library. (MIT)
Netcap	1.8k	2018	2026-10-26 ✅	— security audit framework over network traffic: sensors, audit and collection from live or captured streams. Used by researchers analyzing network captures as structured data; the open-source framework for network-audit work. (GPL-3.0)
go-audit	1.7k	2016	2026-10-21 ✅	— auditd in JSON for Linux: structured events that pipelines can actually parse. Used by defenders replacing auditd for reliable Linux auditing; the open-source audit daemon for scale. (MIT)
Cortex	1.6k	2017	2026-10-20 ✅	— Observable analysis and active response engine pairing with TheHive for automated enrichment of IPs, domains, and hashes. Used by SOC and IR teams; a powerful open-source analysis backend. (AGPL-3.0)
Conpot	1.5k	2013	2026-10-31 ✅	— ICS/SCADA honeypot: emulates the protocols that run factories, safely. Used by researchers deploying ICS/SCADA honeypots; the open-source deception tool for industrial protocols. (GPL-2.0)
Bulk Extractor	1.4k	2012	2026-10-12 ✅	— Extracts email addresses, URLs, and other artifacts from raw data and disk images. Used by forensic investigators; a fast free bulk artifact scanner.
NoScript	1.2k	2018	2026-10-27 ✅	— the browser extension that blocks JavaScript and plugin content by default — client-side hardening's origin story. Used by privacy-focused users blocking scripts by default; the long-standing open-source browser-hardening extension. (GPL-3.0)
ElastAlert2	1.1k	2020	2026-10-27 ✅	— the maintained successor to ElastAlert: flexible alerting on Elasticsearch data with dozens of rule types. Used by SOC teams alerting from Elasticsearch data; the open-source alerting layer for ELK stacks.
AVML (Acquire Volatile Memory f…	1.1k	2019	2026-10-27 ✅	— Microsoft's static binary for acquiring volatile memory on Linux systems during incident response. Used by DFIR practitioners; a dependable free acquisition utility. (MIT)
FirmAE	962	2020	2026-06-24 ⏳	— Automated firmware emulation platform that boots extracted Linux-based IoT firmware in QEMU for dynamic testing. Used by vulnerability researchers; a leading emulation framework. (MIT)
Krawl	826	2025	2026-10-31 ✅	— deception-driven defense: a honeypot server framework for research. Used by defenders discovering and inventorying web assets; the open-source exposure-mapping crawler. (MIT)
HijackThis	818	2016	2026-07-02	— the veteran Windows hijack analyzer: scans browser and OS settings for malicious tampering. Used by technicians diagnosing Windows malware persistence; the veteran free scanner still trusted for autostart analysis. (GPL-2.0)
Hashdeep	791	2012	2025-09-20 ⏳	— File hashing tool with recursive hashing and audit mode for verifying evidence integrity. Used by forensic examiners; a dependable free hashing utility.
RegRipper	725	2020	2026-05-27 ⏳	— Windows registry analysis tool for parsing and extracting forensic artifacts from hive files. Used by incident responders; a widely relied-upon free registry parser.
RITA	667	2024	2026-10-30 ✅	— detects C2 beaconing in Zeek logs — traffic analysis that finds the quiet intruders. Used by threat hunters finding C2 beaconing in DNS and flows; the free open-source hunter for network telemetry. (GPL-3.0)
malware-jail	480	2016	2023-06-16 ⏳	— Sandbox for semi-automatic JavaScript malware analysis, deobfuscation and payload extraction. Used by malware analysts and DFIR teams; a practical JS analysis environment. (MIT)
Sublime Rules	381	2021	2026-11-01 ✅	— the community rule corpus for the Sublime platform. Used by email defenders detecting phishing patterns; the open detection ruleset for the Sublime platform. (MIT)
HEARTH	359	2024	2026-11-01 ✅	— Community-driven threat-hunting repository: 300+ searchable hunting hypotheses and methodologies with detection-coverage heatmaps, built for hunters to share and collaborate. Used by threat hunters looking for their next hypothesis; the open alternative to closed hunt-playbook libraries.. (MIT per README/package.json)

Digital Forensics Framework (DFF)	318	2015	2020-02-13 ⏳	— Open-source platform for digital forensics with a modular analysis framework. Used by forensic examiners; a free extensible investigation environment. (GPL-2.0)
Sublime Platform	277	2021	2026-10-30 ✅	— open email threat detection: rules-as-code against phishing and BEC. Used by email-security teams defending mail flows; the open-source email-defense platform alternative to commercial email gateways. (MIT)
TRACE Forensic Toolkit	228	2023	2026-10-31 ✅	— all-in-one local DFIR desktop app: read-only analysis of E01/AFF4/raw images across a dozen filesystems, BitLocker/FileVault/LUKS unlock, 70-type carving, timelines and hash-chained custody. Used by forensic examiners and students working cases without commercial licenses; the open DFIR suite for evidence-grade analysis. (MIT)
masscanned	140	2021	2026-06-17 ⏳	— Low-interaction honeypot focused on network scanners and bots, integrating with IVRE for self-hosted threat intelligence. Used by blue teams and threat researchers; a lightweight sensor for capturing scanning activity. (GPL-3.0)
logarc	4	2026	2026-11-01 ✅	— offline access-log forensics that reconstructs attacker activity graphs. Used by incident responders analyzing web logs without cloud tooling; the offline log-forensics analyzer. (MIT)
SentinelOps	1	2026	2026-04-06 ⏳	— Lightweight defensive platform ingesting system and network logs to detect brute-force attempts, anomalous logins, and misuse patterns. Used by SOC analysts; a simple detection dashboard.
S.I.D.E. (SCADA Intrusion Detectio…	0	2025	2025-11-09 ⏳	— Intrusion detection system designed for ICS/SCADA networks, monitoring devices, detecting anomalies, and providing real-time alerts. Used by OT defenders; a focused blue-team monitor for industrial traffic. (MIT)
DoD Blue Team Network Lab	0	2025	2025-10-23 ⏳	— Virtualized blue team lab simulating DoD-style enterprise defense with pfSense firewall, VLAN segmentation, and Windows Server. Used by defensive security learners; a hands-on training environment.
Sysinternals Suite	—	—	—	— Microsoft's free Windows admin and DFIR toolkit: Process Explorer, Autoruns, Procmon and dozens more. Used by every Windows admin and IR analyst; Microsoft's free toolkit commercial IR teams also rely on. (free, proprietary)
Hayabusa	3428	2020	2026-10-28 ✅	— Windows EVTX analysis + Sigma detection. (AGPL-3.0)
Nessus	—	—	—	— Vulnerability assessment and security scanning platform from Tenable. Used by defenders and sysadmins; Nessus Essentials is the long-standing free tier for home scanning. (free tier)
Snort	—	—	—	— Classic signature-based intrusion detection system widely used for real-time traffic analysis and packet logging. Used by network defenders and SOC teams; a long-standing staple of open-source IDS deployments. (GPL-2.0)
REMnux	—	—	—	— Linux toolkit for reverse-engineering and analyzing malicious software, providing a curated collection of free community tools in a preconfigured environment. Used by malware analysts and incident responders; a standard Linux analysis distro. (free, proprietary)
Process Monitor (Procmon)	—	—	—	— Sysinternals tool monitoring real-time file system, registry, and process/thread activity for dynamic analysis. Used by malware analysts and incident responders; a core Windows behavioral observation utility. (free, proprietary)
Process Hacker	—	—	—	— Advanced Windows process viewer and memory editor offering detailed inspection of processes, threads, memory regions, and handles. Used by malware analysts and reverse engineers; a powerful live-system inspection tool. (free, proprietary)
Volatility	—	—	—	— Memory forensics framework extracting artifacts from RAM dumps, including hidden processes, injected code, and network connections. Used by DFIR analysts and forensic investigators; the leading memory analysis framework. (free, proprietary)
PEStudio	—	—	—	— Static analysis tool for Windows executables examining PE headers, imports, exports, strings, and indicators without execution. Used by malware analysts and triagers; a fast first-pass PE inspection utility. (free, proprietary)
Any.run	—	—	—	— Interactive online sandbox with real-time analysis letting analysts observe and interact with malware execution. Used by malware analysts and researchers; a popular cloud sandbox with a genuine free tier. (free tier)
Hybrid Analysis	—	—	—	— Public sandbox powered by Falcon Sandbox providing automated malware analysis with detailed behavioral reports. Used by analysts and researchers; a widely used free cloud analysis service. (free tier)

github.com/RanaKhan77/awesome-cyber-security-heaven - CC0 - Powered by Awesome Cloud Analysis Service

RedSentinel	—	—	—	— AI-assisted defensive security assessment framework orchestrating Nmap, Nikto, and WhatWeb, normalizing output and enriching findings. Used by blue teamers; a defensive scanning orchestrator.
Hayabusa Lens	—	—	—	— Desktop application wrapping Hayabusa and Chainsaw with a dashboard, Sigma rule viewer, and AI investigation helpers. Used by DFIR analysts; a convenient Windows event triage frontend.
Graylog	—	—	—	— Centralizes logs and provides alerting and dashboards through a polished interface. Used by log analysts; a capable open-source log management platform. (SSPL)
Elastic Stack (ELK)	—	—	—	— Infrastructure for log storage, processing, and visualization rather than a complete SIEM, with Elasticsearch and Kibana relicensed. Used by log analysts; a widely adopted open log stack.
OpenSearch	—	—	—	— Fork of Elasticsearch and Kibana led by AWS, including OpenSearch and OpenSearch Dashboards for visualization. Used by log analysts; a truly open search and analytics stack. (Apache-2.0)
SecurityOnion	—	—	—	— Functions as both a SIEM and intrusion detection system, integrating Snort, Suricata, and Wazuh for defense. Used by blue teams; a comprehensive open-source security platform.
TestDisk	—	—	—	— Disk recovery tool that restores lost partitions and repairs filesystem structures. Used by forensics and recovery technicians; a reliable free utility.
PhotoRec	—	—	—	— File recovery software that carves deleted files from disks and memory cards. Used by forensics examiners and recovery specialists; a free companion to TestDisk.
Foremost	—	—	—	— File carving tool that recovers data from disk images based on file headers. Used by forensic investigators; a classic free carving utility.
Pdf-parser	—	—	—	— Didier Stevens' PDF analysis tools for extracting and examining content from suspicious PDF files. Used by malware analysts; a trusted free PDF forensics toolkit.
PEView	—	—	—	— Portable executable file viewer for inspecting PE headers and structure during malware analysis. Used by reverse engineers; a simple free static analysis viewer.
AIDE (Advanced Intrusion Detecti…	—	—	—	— File integrity monitoring tool creating cryptographic checksums of system files and detecting unauthorized modifications. Used by blue teams; a classic host-based integrity baseline. (GPL-2.0)
The DFIR Report	—	—	—	— Real-world incident response case studies with detailed timelines, IOCs, and attacker TTPs. Used by blue teamers and threat hunters; an essential reference for detection and response.
CAINE	—	—	—	— Digital forensics live Linux distribution. Used by forensic investigators; a ready-to-run environment for evidence analysis.
SELKS	—	—	—	— Suricata-based IDS/IPS platform for network detection. Used by defenders and SOC analysts; an open-source intrusion detection stack.